

- Include a callback URL (see [Section 5.7.3.1, “CommissioningCustomFlowUrl format”](#) for **MTcu** key details) within the **ExpandedFallbackUrl**.
- The Manufacturer Website or App MAY utilize the **CallbackUrl** field, if provided in the query string, in order to simplify the process for signaling the completion of the manufacturer-specific part of the flow back to the Commissioner. When doing so, the Manufacturer Website or App SHOULD put the device into Commissioning mode and SHOULD provide the corresponding onboarding payload to the Commissioner using the **MTrop** key/value pair within the **Expanded-CallbackUrl**.

5.7.6. Onboarding Payload Inclusion (Manual Pairing Code, QR Code, NFC Tag)

The Onboarding Payload formats (Manual Pairing Code, QR code, or NFC Tag) enable secure commissioning and provide a consistent experience that many users are familiar with. However, because they contain a symmetric security code, it is not appropriate in all circumstances to have them be in a readily accessible location on the device, such as printed on the back.

The following are the requirements and recommendations regarding the QR Code, NFC Tag and Manual Pairing code for Standard and User Intent Commissioning Flow Devices, as well as for Custom Commissioning Flow Devices when they satisfy the conditions described in [Section 5.7.3, “Custom Commissioning Flow”](#).

The term 'on-device' allows for a physical label affixed to the device or printed directly on the device, as well as one that can be displayed on demand through some physical interface properties of the device (e.g. visual or audio).

1. Devices SHALL include the Manual Pairing code on-device or in packaging.
2. Devices SHALL NOT have the QR nor the manual pairing code in an unprotected format on the outer packaging.
3. Devices SHOULD include the QR Code, and SHOULD include it alongside the Manual Pairing Code on-device or in packaging.
 - a. Permutations (e.g. Manual Pairing Code on-device and QR Code in packaging) are allowed.
4. Manual Pairing Code and QR Code on-device MAY be removable or obscured to allow the owner to prevent commissioning without their consent.
5. Devices MAY include the QR Code and Manual Pairing Code in multiple forms (see below).
6. Devices MAY include an NFC Tag.
7. Devices including an NFC Tag SHOULD include it alongside the mandatory Manual Pairing Code on-device or in packaging.
 - a. Permutations (e.g., an NFC Tag on-device and a Manual Pairing Code in packaging, or vice versa) are allowed.
8. An on-device NFC Tag MAY be removable or disabled to allow the owner to prevent commissioning without their consent.

Presentation of the QR Code and Manual Pairing code on-device can occur in many forms to allow for adherence to device security requirements and manufacturing considerations. For example